Task1:

```
PS C:\Users\QT\Desktop\DSAlabttasks> cd "c:\Users\QT\Desktop\DS
Enter number of elements: 5
Enter elements from bottom to top:
0
1
2
3
4
Maximum value: 4
Minimum value: 0
Middle element: 2
PS C:\Users\QT\Desktop\DSAlabttasks\labtask5>
```

Task2:

```
PS C:\Users\QT\Desktop\DSAlabttasks> cd "c:\Users\QT\Deskto
1
2
3
Enter size of Stack 2: 3
Enter elements of Stack 2 from bottom to top:
2
3
4

Third stack before sorting: 4 3 3 2 2 1
Third stack after sorting: 1 2 2 3 3 4
Circular Queue: 1 2 2 3 3 4
PS C:\Users\QT\Desktop\DSAlabttasks\labtask5>
```

Task3:

```
Expression 1:
Postfix: 1 c o s t h e t a - 1 c o s t h e t a + /
Prefix:   / - 1 c o s t h e t a + 1 c o s t h e t a

Expression 2:
Postfix: 1 c o s t h e t a 2 ^ -
Prefix:   - 1 ^ c o s t h e t a 2

Expression 3:
Postfix: 1 2 / c o s a l p h a b e t a - c o s a l p h a b e t a + - *
Prefix:   * / 1 2 - c o s - a l p h a b e t a c o s + a l p h a b e t a
PS C:\Users\QT\Desktop\DSAlabttasks\labtask5>
```

Task4:

```
PS C:\Users\QT\Desktop\DSAlabttasks> cd "c:\Users\QT\Desktop\DSAlabttasks\labtask5\" ; if ($?) { gcc task4.c
Enter Expression 1: x^2 + 2*x + 1
Enter Condition 1: x>=0
Enter Expression 2:  2*x^2 - x + 3
Enter Condition 2: x<0
Enter x: 5

The program selects:
Condition: x>=0
Expression: x^2 + 2*x + 1
Postfix: x2^2x*+1+
Calculated Value: 36.00
PS C:\Users\QT\Desktop\DSAlabttasks\labtask5>
```

Task5:

```
PS C:\Users\QT\Desktop\DSAlabttasks> cd "c:\Users\QT\Desktop
Enter number of terms: 3
Series: 1 + 4 + 7
Postfix: 1 4 + 7 +
Prefix: 1 4 7 + +
PS C:\Users\QT\Desktop\DSAlabttasks\labtask5>
```